Probability of a lysosome missing a phagosome

07/01

Lysosome : ρ_l (density)
v_l (speed)
k_l (production rate)

Assuming a fusion probability $p_f = 1$

Phagosome : ρ_g, v_g, k_g

$$p(t) = \left(1 - \int_0^{\frac{L}{v_g} + t + L/v_p} k_g e^{-k_g t'} dt'\right) \left(\int_0^{t} k_l e^{-k_l t'} dt'\right)$$

p(no phagosome produced by the time previous one reaches the end, new lysosome is produced and reaches the other end)

p(lysosome is produced in time t)

$$= \left(\exp\left(-k_g\left(\frac{L}{v_g} + t + \frac{L}{v_p}\right)\right)\right)\left(1 - \exp\left(-k_l t\right)\right)$$

Phagosome distance : $1/\rho_g \sim 62.5$ microns